

Generation Interconnection System Impact Study Report

For Queue Project Q22-1992

Uintah 138 kV
November 29, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-1992. The Interconnection Customer proposes a 54.2 MW battery energy storage facility interconnecting at the Uintah 138 kV substation in the Virginia area. The request was received on January 31, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 508 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$351,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

- Facility: Shuffleton - Arvin 138 KV line
Contingency: Kearneyville - Clark County 345 KV
Loading: 111.3% (Rating: 277 MVA)
DFAX: 6.2%
Pre-project loading: 110.1%
- Facility: Douglas City - Southwest Research Institute 138 KV
Contingency: Garen Corners - Douglas City Substation 138 KV
Loading: 100.5% (Rating: 291 MVA)
DFAX: 6.0%
Pre-project loading: 99.3%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Arvin - Shuffleton 138 KV	Rebuild line with higher-capacity conductor	\$4,299,000	5% (\$210,000)
Douglas City - Southwest Research Institute 138 KV	Reconductor line	\$1,225,000	12% (\$141,000)
	Total allocated to project		\$351,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.